

Sirui Shen

PH.D. STUDENT

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Education

Centrum Wiskunde & Informatica

Amsterdam, Netherlands

PH.D. STUDENT IN COMPUTER SCIENCE

Sep. 2023 - Present

- Research topic: Hardware Security, Garbled Circuit, Zero-knowledge Proof, Side-channel Analysis.
- Current project: Zero-knowledge proof for formal verification.
- Advisor: Dr. Chenglu Jin, Prof. Marten van Dijk
- Affiliated with the Vrije Universiteit Amsterdam

Nanjing University

Nanjing, China

MASTER IN ELECTRONIC SCIENCE AND ENGINEERING

Sep. 2020 - Jun. 2023

- Thesis: Hardware Design and Implementation of Core Operators in Lattice-Based Cryptography.

Nanjing University

Nanjing, China

BACHELOR IN ELECTRONIC SCIENCE AND ENGINEERING

Sep. 2016 - Jun. 2020

- Major in Microelectronic Science and Engineering.

Publication

1. S. Shen, & Z. Huang, & C. Jin (2026). Proving Circuit Functional Equivalence in Zero Knowledge. Accepted by the 2026 ACM Conference on Computer and Communications Security (CCS).
2. S. Shen, & C. Jin (2024). Reading It like an Open Book: Single-trace Blind Side-channel Attacks on Garbled Circuit Frameworks. In Proceedings of the Annual Computer Security Applications Conference (ACSAC), 410-424
3. H., Zhang, S., Shen, X. Hu, & C., Jin (2025). Ransomware Negotiation: Dynamics and Privacy-Preserving Mechanism Design. In International Conference on Game Theory and AI for Security (GameSec), 235-255.
4. W. Song, S. Shen, C. Xu, Y. Wang, X. Wang, Y. Fu, ... & Z. Lu (2024). Heterogeneous Reconfigurable Accelerator for Homomorphic Evaluation on Encrypted Data. IEEE Access.
5. Y. Fu, K. cheng, W. Song, G. He, S. Shen, H. Wang, ... & L. Li (2022). A DSP-purposed Reconfigurable Acceleration Machine (DREAM) for High Energy Efficiency MIMO Signal Processing. IEEE Transactions on Circuits and Systems I: Regular Papers, 70(2), 952-965.
6. S. Shen, W. Song, X. Wang, X. Shao, Y. Fu, Z. Lu, and L. Li. A Hierarchical Parallel Discrete Gaussian Sampler for Lattice-based Cryptography. In 2022 IEEE International Symposium on Circuits and Systems (ISCAS), 1-5.
7. Q. Chen, G. He, X. Wang, J. Xu, S. Shen, H. Chen, ... & L. Li (2021). A 67.5 μ J/prediction accelerator for spiking neural networks in image segmentation. IEEE Transactions on Circuits and Systems II: Express Briefs, 69(2), 574-578.

Service

Reviewer

Conferences

- IEEE International Symposium on Circuits and Systems (ISCAS'26)
- IEEE International Symposium on Circuits and Systems (ISCAS'24)

Artifact Evaluation

- The annual Conference on Cryptographic Hardware and Embedded Systems (CHES’26)

Sub-reviewer

Journals

- IEEE Transactions on Information Forensics and Security
- ACM Transactions on Design Automation of Electronic Systems

Conferences

- The annual Conference on Cryptographic Hardware and Embedded Systems (CHES’26)
- The 24th Workshop on Privacy in the Electronic Society (WPES’25)
- Financial Cryptography and Data Security 2025 (FC’25)
- The annual Conference on Cryptographic Hardware and Embedded Systems (CHES’25)
- The 26th International Symposium on Quality Electronic Design (ISQED’25)
- Workshop on Attacks and Solutions in Hardware Security (ASHES’24)
- IEEE Conference on Communications and Network Security (CNS’24)
- The 25th International Symposium on Quality Electronic Design (ISQED’24)
- International Workshop on Security and Trust Management (STM’24)
- IEEE International Symposium on Circuits and Systems (ISCAS’23)

Honors & Awards

2025	Runner-up , Dutch Cyber Security Best Research Paper Award Technical Track	ACCSS
2024	Student Travel Stipend , Real World Crypto	IACR
2022	Scholarship , National Scholarship	Nanjing University
2022	Champion , National Graduate Electronic Design Contest, Security Track	Fudan Micro Co., Ltd.
2021	The First Class , Postgraduate Academic Scholarship	Nanjing University
2018	Third Prize , People’s Scholarship	Nanjing University
2017	Scholarship , Gentrue Scholarship	Genture Electronics Co., Ltd.